

# Docker Multi-Stage Builds

In multistage builds, we write multiple **FROM** statements in one **Dockerfile**. Each one starts a fresh image. You copy only what you need from earlier stages into the final one.

**Multistage** build is all about **optimization (reducing image size)**

## Why it exists:

The build environment is bloated. (Compilers, dev dependencies, source code, etc.)  
None of them is needed in production.

Multistage lets us:

- Build in the first stage
- Just pick the compiled binary into the final image.  
(leads into shorter image size.)

## Example:

For containerizing a React app,

We can:

- Build it in one stage and
- Just copy the dist folder (Compiled output of our source code, ready for production) in the last stage

```
# STAGE 1: BUILD
FROM node:22-alpine AS builder
WORKDIR /app
COPY . .
RUN npm ci && npm run build

# STAGE 2: RUN
FROM node:22-alpine
WORKDIR /app
COPY --from=builder /app/dist ./dist
CMD ["node", "dist/index.js"]
```

**Stage 1** has all the build tools. **Stage 2** only gets the output. The final image is small and clean. (Build heavy, ship light)